

Alex Carter

AI Engineer & Researcher

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SUMMARY

AI engineer with 6+ years turning research into revenue - deployed LLM platforms serving 2M+ users, cut inference costs by 70% through model compression, and published at NeurIPS and EMNLP. Full-stack ML: data pipelines, training infrastructure, evaluation, and reliable deployment.

EXPERIENCE

Senior AI Engineer - Nexus AI

2022 - Present

- Led the LLM platform team (5 engineers) building RAG infrastructure serving 2M+ monthly queries at 94% answer accuracy.
- Cut inference spend 70% (\$336k/yr) by shipping distilled domain models with automated eval gating.
- Designed the company-wide LLM evaluation framework now used by 8 product teams.

Machine Learning Engineer - Vantage Robotics

2020 - 2022

- Shipped real-time edge CV models achieving 99.2% defect recall; drove \$1.4M annual savings for the flagship client.
- Built the active-learning data engine that cut labeling costs 60% while improving mAP by 11 points.

Research Engineer - Stanford AI Lab (SAIL)

2018 - 2020

- Co-authored 2 publications on efficient deep learning (NeurIPS workshop, EMNLP).
- Built distributed training tooling on 64-GPU clusters, reducing experiment turnaround from days to hours.

SELECTED PROJECTS

RAGStack - Production RAG Platform

Hybrid retrieval + reranking, streaming via vLLM, eval-gated deploys. 94% answer accuracy, 800ms p95, 2M+ queries/month.

VisionEdge - Real-time Defect Detection

YOLOv8 + TensorRT INT8 on edge GPUs with OTA updates. 99.2% recall, 18ms/frame, \$1.4M annual savings.

DistillLab - LLM Compression Research

Distilled 70B teacher into 7B student via rejection-sampled synthetic data + LoRA. 70% cost cut, 98.1% quality retained (EMNLP 2024).

AtlasOps - ML CI/CD Platform

Feature store, Ray training, canary deploys, drift monitoring via GitOps. Deploy time 6 weeks to 4 hours; 40+ models in production.

PUBLICATIONS

Task-Aware Distillation: Retaining 98% of Teacher Quality at 10% of the Cost - EMNLP 2024 (87 citations)

Benchmarking Retrieval Robustness in Production RAG Systems - NeurIPS Workshop on Efficient NLP, 2023 (142 citations)

SKILLS

Languages: Python, TypeScript, Go, SQL, C++, Rust

ML/DL: PyTorch, JAX, TensorFlow, scikit-learn, XGBoost, Lightning

LLM Tooling: HuggingFace, LangChain, vLLM, OpenAI API, Pinecone, pgvector, LlamaIndex

MLOps/Infra: Docker, Kubernetes, AWS, GCP, Terraform, MLflow, Ray, Airflow

Data: Spark, Kafka, dbt, PostgreSQL, Redis, BigQuery